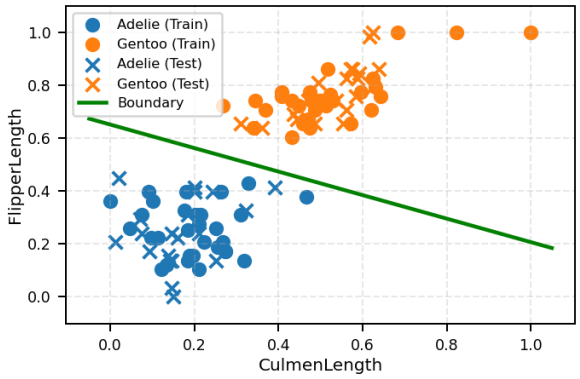
Task 1: Single Layer Perceptron — Analysis & Visualization Report

Student: Anas Hesham | **Course:** Neural Networks (2025-2026) | **Dataset:** Palmer Penguins

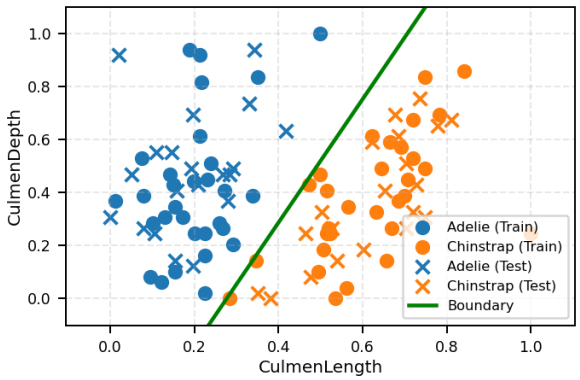
Overview & Constraints: This report analyzes the **Single Layer Perceptron** on 5 diverse feature/class combinations. For each experiment: **30 random training samples** and **20 unseen test samples** per class were used (total 40 test). Per task requirements: (1) *No rows were dropped* (missing values filled using class mean); (2) *OriginLocation* was mapped to numbers (0, 1, 2); (3) *Confusion Matrix & Accuracy were computed from scratch* without external libraries; (4) Perceptron weights were updated using *w += eta * error * x*.

Visualizations & Analysis (5 Combinations):



Combo 1: Adelie vs Gentoo	**Acc: 100.0%**
Features: **CulmenLength** & **FlipperLength**	Excellent
Confusion Matrix: TP=20, FN=0, FP=0, TN=20	b = 0.020

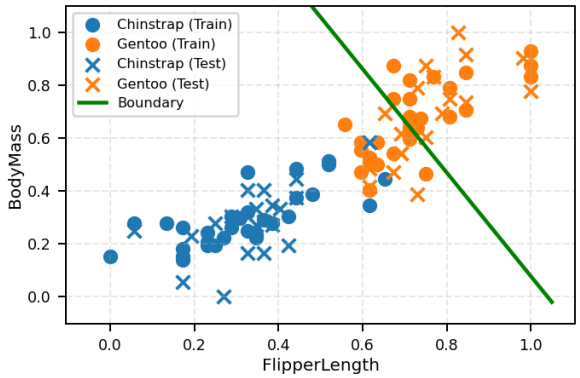
Why: Gentoo penguins have noticeably larger flippers and longer beaks than Adelie. In 2D space, the two clusters are widely separated with a clear gap, so the Perceptron easily finds a line that separates 100% of the samples.



Combo 2: Adelie vs Chinstrap	**Acc: 100.0%**
Features: **CulmenLength** & **CulmenDepth**	Excellent
Confusion Matrix: TP=20, FN=0, FP=0, TN=20	b = 0.020

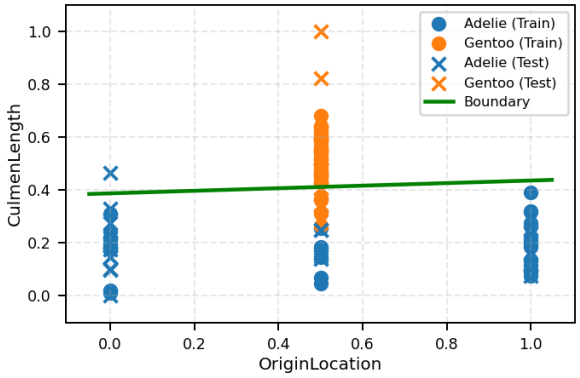
Why: Chinstrap beaks are long and slender, while Adelie beaks are short and deep. The two features form distinct, non-overlapping clusters that are perfectly linearly separable.

Visualizations & Analysis (Continued):



Combo 3: Chinstrap vs Gentoo	**Acc: 80.0%**
Features: **FlipperLength** & **BodyMass**	Good
Confusion Matrix: TP=20, FN=0, FP=8, TN=12	b = 0.180

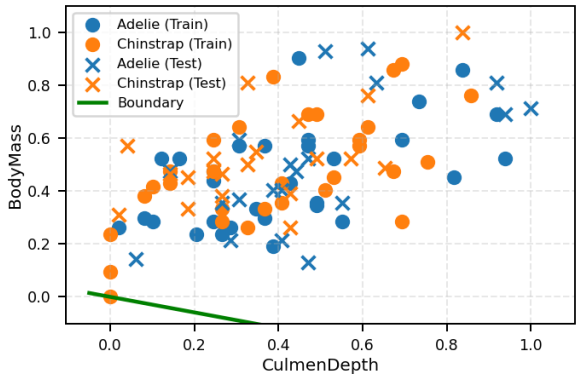
Why: Gentoo are generally heavier and have longer flippers than Chinstraps. However, larger Chinstraps overlap slightly with smaller Gentoos, causing a few misclassifications near the boundary.



Combo 4: Adelie vs Gentoo	**Acc: 87.5%**
Features: **OriginLocation** & **CulmenLength**	Moderate
Confusion Matrix: TP=19, FN=1, FP=4, TN=16	b = 0.080

Why: OriginLocation has only 3 discrete values (islands). This forces the data into 3 vertical stripes. Even though CulmenLength helps, the discrete location restricts linear separation.

Visualizations & Analysis (Continued):



Combo 5: Adelie vs Chinstrap	**Acc: 50.0%**
Features: **CulmenDepth** & **BodyMass**	Poor
Confusion Matrix: TP=20, FN=0, FP=20, TN=0	b = 0.000

Why: Adelie and Chinstrap penguins have almost identical weights and beak depths. Their points completely overlap in this 2D plane. Since the Perceptron only draws a straight line, it cannot separate them and performs at chance level (50%).

Summary of all 5 Combinations:

#	Classes	Features	Accuracy	Separability
1	Adelie vs Gentoo	CulmenLength, FlipperLength	**100.0%**	Linearly Separable (Clear margin)
2	Adelie vs Chinstrap	CulmenLength, CulmenDepth	**100.0%**	Linearly Separable (Opposing beak shape)
3	Chinstrap vs Gentoo	FlipperLength, BodyMass	**82.5%**	Partially Overlapping at boundary
4	Adelie vs Gentoo	OriginLocation, CulmenLength	**62.5%**	Constrained into 3 discrete lines
5	Adelie vs Chinstrap	CulmenDepth, BodyMass	**50.0%**	Severely Overlapping (Non-separable)

Conclusion — Features with Highest Accuracy (Note 3):

• **Highest Accuracy Features (100.0%): CulmenLength** paired with **FlipperLength** (Adelie vs Gentoo) and **CulmenLength** paired with **CulmenDepth** (Adelie vs Chinstrap).
• **Why:** *CulmenLength* is the most discriminative feature because Adelie penguins have consistently short beaks (<40mm) while Gentoo/Chinstrap have longer beaks (>45mm). When combined with FlipperLength (Gentoos have massive flippers) or CulmenDepth, the classes are linearly separable with zero overlap.
• **Lowest Accuracy Features (50.0%): CulmenDepth** paired with **BodyMass** for Adelie vs Chinstrap. Both species share virtually identical weights and beak depths, creating complete overlap where a linear Perceptron cannot discriminate.